Carboxylic acids and derivatives

A Level Chemistry

Carboxylic acids

Making and reacting

- an **alkylbenzene** 烷基苯 (such as methylbenzene) is oxidised by hot alkaline $KMnO_4$, then dilute acid, to give **benzoic acid** 苯甲酸. The whole side-chain becomes a –COOH group.
- a carboxylic acid reacts with PCl_3 and heat, PCl_5, or $SOCl_2$ to form an **acyl chloride** 酰氯.

Acids that can be oxidised further

Two **carboxylic acids** 羧酸 are special because they can still be oxidised:

- methanoic acid (HCOOH) is oxidised by Fehling's or Tollens' reagent, or acidified $KMnO_4$ / $K_2Cr_2O_7$, to carbon dioxide and water.
- ethanedioic acid (HOOCCOOH) is oxidised by warm acidified $KMnO_4$ to carbon dioxide.

Relative acidities

The **acidity** 酸性 order is:

$$\text{alcohol} < \text{phenol} < \text{carboxylic acid}$$

A carboxylic acid is the strongest because, when it loses H^+, the negative charge is spread over **two** oxygen atoms, making the ion very stable. In a **phenol** 苯酚 the charge spreads only into the ring, and in an **alcohol** 醇 (giving RO^-) it is not spread at all.

Chlorine atoms make a carboxylic acid **more** acidic. Chlorine is **electron-withdrawing** 吸电子: through the **inductive effect** 诱导效应 it pulls electron density away, helping to spread the negative charge and stabilise the ion. So more chlorine atoms (closer to the –COOH) give a stronger acid.

Esters

An alcohol (or phenol) reacts with an acyl chloride at room temperature to give an **ester** 酯 and HCl. Examples are ethyl ethanoate (from ethanol) and phenyl benzoate (from phenol).

Acyl chlorides

Acyl chlorides are made from carboxylic acids (with PCl_3, PCl_5 or $SOCl_2$). They are very reactive. At room temperature they react with:

Reactant	**Product (plus HCl)**
water	the carboxylic acid
an alcohol	an ester
phenol	an ester
ammonia 氨	an **amide** 酰胺
a primary or secondary **amine** 胺	an amide

The addition–elimination mechanism

All these reactions follow an **addition–elimination** 加成消去 mechanism: a nucleophile first **adds** to the slightly positive carbonyl carbon, then HCl is **eliminated**.

Ease of hydrolysis

Compare how easily three chlorides react with water (**hydrolysis** 水解):

$$\text{acyl chloride} \gg \text{alkyl chloride} \gg \text{aryl chloride}$$

An acyl chloride reacts violently with cold water; an alkyl chloride reacts slowly; an aryl chloride (halogenoarene) does not react, because its C–Cl bond is strengthened by the ring.